

Module 06 [Data Structure with STL] (for Week 7)

Topic 1 [STL: stack class]

Problem statement: Write a menu program to create and manipulate Stack using `stack` class and perform the following operations using the specified method.

- i) use `push()` method to push data
- ii) use `pop()` method to pop data
- iii) use `top()` method to display top element
- iv) use `empty()` method to check whether stack is empty or not

```
** Stack **  
1. Push  
2. Pop  
3. Display()  
4. Exit  
Enter your option:
```

Topic 2 [STL: queue class]

Problem statement: Write a menu program to create and manipulate Queue using `queue` class. Perform the following operations using the specified method.

- i) use `push()` method to enqueue data
- ii) use `pop()` method to dequeue data
- iii) use `front()` method to display front element
- iv) use `back()` method to display rear element
- v) use `empty()` method to check whether queue is empty or not

```
** Queue **  
1. Enqueue  
2. Dequeue  
3. Display Front  
4. Display Rear  
5. Exit  
Enter your option:
```

Topic 3 [STL: vector class]

Problem statement: Write a menu program to create and manipulate linked list using `vector` class and use the following methods

- i) **`push_back()`**
- ii) **`push_front()`**
- iii) **`begin()`**
- iv) **`end()`**

- v) **pop_back()**
- vi) **pop_front()**
- vii) **insert()**
- viii) **erase()**

```
** Linked List **  
1. Insert  
2. Delete  
3. Update  
4. Search  
5. Exit  
Enter your option:
```

Topic 3.1 [STL: vector class]: Add more feature to the menu of Topic 3.